Knowledge, concepts, and views

One knowledge model; rival theories of semantic memory and concepts, compared by proof

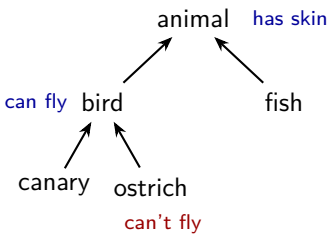
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One knowledge model

- A **knowledge net**: labelled links between nodes; one label is the hierarchy (*isa*).
- Nodes are **terms**: atoms (*canary*), structured values (*set the program-plan to MOVE FORWARD 9*), and typed *holes* (*a Real-World-Action*).
- **Everything is links**: frames and slots, patterns, a concept's prototype (its votes), and even *how* a concept decides membership.
- **Processes are generic**: inheritance, verification, prototype similarity, categorization, view application. A theory supplies data and choices, not a new representation.

Semantic nets: what is proved for every net



- **Inheritance**: a property stored once at a superordinate is found by every subordinate (cognitive economy).
- **Override**: a single-valued property stored nearer wins (an ostrich can't fly).
- **Accumulation**: many-valued properties collect up the hierarchy (a canary has wings *and* skin).
- **Distance**: the least number of *isa* links.

Two theories of semantic verification

Hierarchical network
(Collins & Quillian 1969)

- Move up from the subject, searching properties at each node.
- Each move and each retrieval costs; costs add.
- Cost depends only on *where* things are stored.

Feature comparison
(Smith, Shoben & Rips 1974)

- Stage 1: overall similarity of feature lists; very high, *true*; very low, *false*.
- Otherwise stage 2: compare defining features only.
- Cost depends on *similarity*.

Costs are **ordinal**, steps or stages: the theories disagree about the *order* of verification times, and that is what is proved.

Laws of verification

Stated for *any* verifier, against a hierarchy that says what is true:

- **Distance effect**: the farther category costs more (*a canary is an animal* after *a canary is a bird*).
- **Typicality effect**: two instances equally far below a category, verified at different costs (*robin* faster than *chicken*).
- **Reversal**: an instance verified faster as a member of the farther category (*a chicken is an animal* faster than *a chicken is a bird*).

Proved once: a reversal contradicts the distance effect.

The semantic-memory matrix

	Distance	Typicality	Reversal
Hierarchical (C&Q)	**has**	**never**	**never**
Feature comparison (SSR)	lacks	can hold	can hold
People (RSS 1973; SSR 1974)	mostly	yes	yes

- C&Q's cells hold for *every* net: cost is distance, so equally distant instances always cost the same.
- The feature model shows both effects with bird features; for every setting, a more similar instance is never slower.
- C&Q's Figure 1 gives the six sentence types of Fig. 2 in the paper's order.

Concepts: combination and over-extension

- **Selective modification** (Smith, Osherson, Rips & Keane 1988): *red* shifts the color votes of *apple* to red and boosts color's diagnosticity.
- **Threshold membership** (Hampton 2007): membership is a nondecreasing function of the same similarity.
- Proved for every setting: the **conjunction effect**, hence **over-extension** (more a member of *red apple* than of *apple*).
- The paper's figures reproduced exactly.

Prototypes are links in the knowledge net, so these theorems apply to any concept held there.

The trouble with numbers

- *multiple of 3*, learned from 3, 6, . . . , 30; *even multiple of 3* by modification.
- 14 is an *even multiple of 3* to degree .80, a *multiple of 3* to .06.
- **Incoherent**: more a member of the combination than of its head.
- **False**: Lean proves 14 is not a multiple of 3.

People know better. So for numbers, something replaces similarity as the arbiter of membership, without removing the prototype.

View application (Shrager 1987)

A *view* is an abstract schema from outside the domain. Applying it to a theory:

1. **copy** the view;
2. **merge** the theory: match type frames by abstract type, or at random within a kind; the view wins on matched slots;
3. **coerce** re-parented instances through their old and new parents' patterns;
4. **bind** the view's holes from the theory;
5. **single function**: an object bound to one operation leaves the others.

Random ties, never revisited: application gives a *distribution* over theories. Proved for every view and theory: the view is kept, the view wins on matched frames, unmatched frames survive, instances are re-parented.

Faithfulness: the paper's Table 4

- Tables 1 and 2 of the paper as data: a BigTrak theory and the CODE.AND.STARTUP view.
- Applying the view gives **Table 4 exactly**, with probability 1/2.
- The other half: *Action-operation* matched to *Immediate-operation* instead of *Memory-operation*, the random choice the paper describes.
- *Do (MOVE FORWARD 9)* becomes *Set the program-plan to (MOVE FORWARD 9)* by coercion; GO leaves Sequence-1; the script runs the memory operation, then the immediate one.

All five steps are the library's; BigTrak is only data.

An ARITHMETIC view for number concepts

The learner's theory

- *category*: membership by similarity (threshold); combination by modification.
- *multiple of 3*, *even*: prototypes learned from exemplars.

The ARITHMETIC view

- *category*: membership by divisibility (divisor from the concept's name); combination by conjunction.

Applying it: *category* matches *category*, so the view's test and combination win. The concepts' prototypes are not matched, so they are **kept**.

Before and after the view

	Before	After
14 in *multiple of 3*	.06	0
14 in *even multiple of 3*	.80 (over-extension)	0
even multiple of 3	similarity, modified prototype	**exactly** $6 \mid n$, every n
Over-extension	at 14	**never**, at any n
Typicality		**unchanged**, every n

- **Dissociation**: 14 as an *even multiple of 3* is more typical (-16) than 9 as a *multiple of 3* (-23), yet only 9 is a member. Before the view, impossible: membership is a nondecreasing function of typicality.
- Graded typicality for an all-or-none concept: Armstrong, Gleitman & Gleitman (1983), *odd number*.

Choosing the view: spreading activation

The 1987 paper had no principled account of *which* view to apply. Collins & Loftus (1975) supply one:

- Activation starts at the **cue** (a key's label) and at the **concepts the current theory uses**.
- It spreads along links, each passing on a fraction set by its strength; activation from different sources **sums**.
- A view is chosen in proportion to its activation, then **applied**.

Proved for every net: activation sums over sources; it is never negative; priming never lowers it; priming a candidate never lowers its share.

The composition is a mixture, chance of choosing a view times chance of each result: the shape of recognition-then-retrieval in the multi-digit model.

Same key, different knowledge

A hypothetical device with a key labelled "+":

- "+" suggests CALCULATING ("+" adds to the display), and more weakly PROGRAMMING ("+" appends a step).
- *Memory* suggests PROGRAMMING.

Learner's theory	"+" adds	"+" appends a step
no memory	2/3	1/3
has a memory	2/5	3/5

The same cue brings in different knowledge, depending on what the learner already knows. Strengths are illustrative; the composition is general.

The concept matrix

	Distance	Typicality	Reversal	Conj. effect	Over-ext.	Dissociation
Hierarchical net	**has**	**never**	**never**	–	–	–
Feature comparison	lacks	can hold	can hold	–	–	–
Selective modification	–	–	–	**has**	**has**	**never**
+ ARITHMETIC view	–	–	–	**has**	**never**	can hold
People	mostly	yes	yes	yes	yes	yes

- **has**/**never**: every setting (conditions stated). lacks/can hold: one setting. –: not stated for that theory.
- Without a view, membership never goes against typicality (any nondecreasing threshold); with it, over-extension never occurs for number concepts.

What this adds, and next

- **One model, many theories**: semantic memory, concept combination and theory change share nets, terms and processes; each theory is data and choices.
- **A rival-theory matrix for knowledge**, as for memory.
- **View application, formalized**: the paper's example exactly, a new consequence for concepts, and view selection by spreading activation.

Next:

- View selection by EPAM recognition of cues, as a rival to spreading activation; garden paths and their repair.
- Collins & Loftus's decision process, making spreading activation a third verification family; feature lists as net links.
- Learning views; a cognitive-skills package (problem solving).